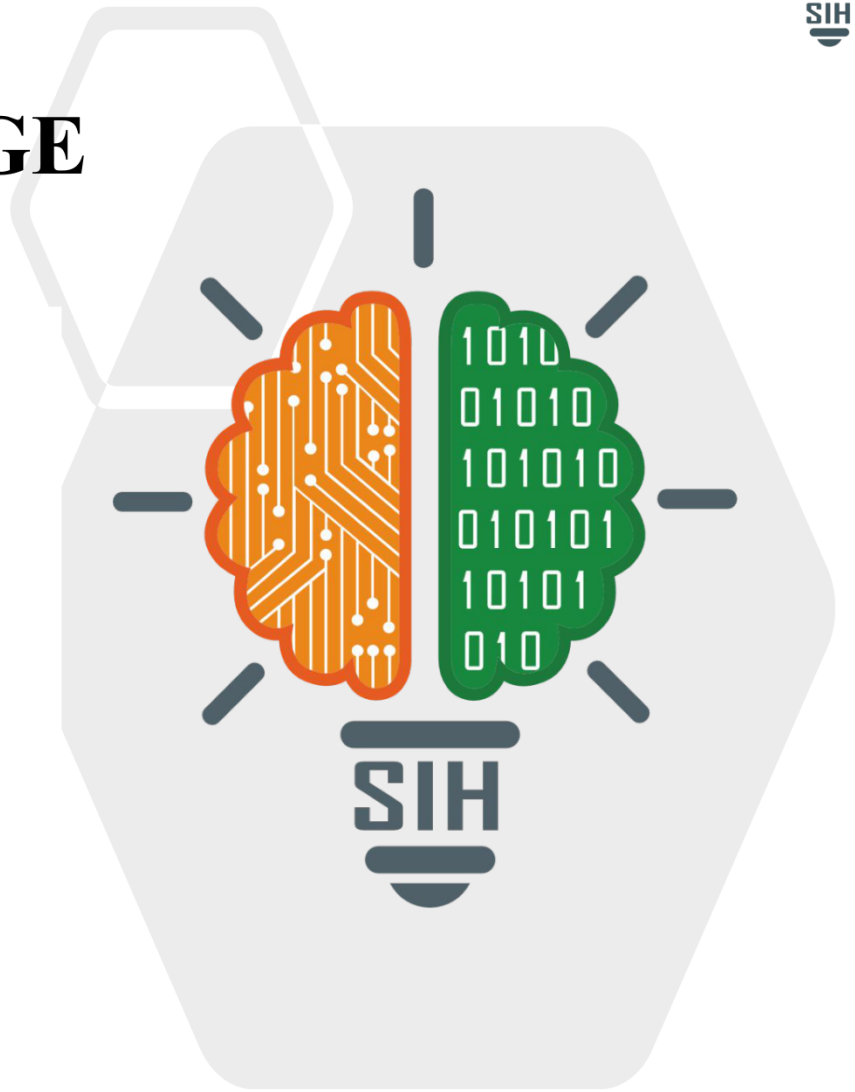
TITLE PAGE



- **Problem Statement ID –** [PS ID]
- **Problem Statement Title-** APAAR-based Student Data Management and Verification System for AICTE-approved Institutions
- **Theme-** [Theme]
- **PS Category-** Software
- **Team ID-** [Team ID]
- **Team Name (Registered on portal)-** [Team Name]

Proposed solution: EDUVERIFY

Team Name

EDUVERIFY

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PROPOSED SOLUTION

Institutions upload **APAAR-linked student records** after admissions. EDUVERIFY maps each file to a standard schema, checks fields, consistency and **category-wise intake limits**, flags **possible duplicates across institutions** with reasons, and routes unclear cases to **authorised reviewers**. Every decision is audit-logged.

Validation ≠ official verification. Our checks confirm records are complete and consistent. Confirming an APAAR ID with the official registry needs authorised API access; the prototype uses a mock registry and marks records "mock-checked".

WHAT MAKES IT DIFFERENT

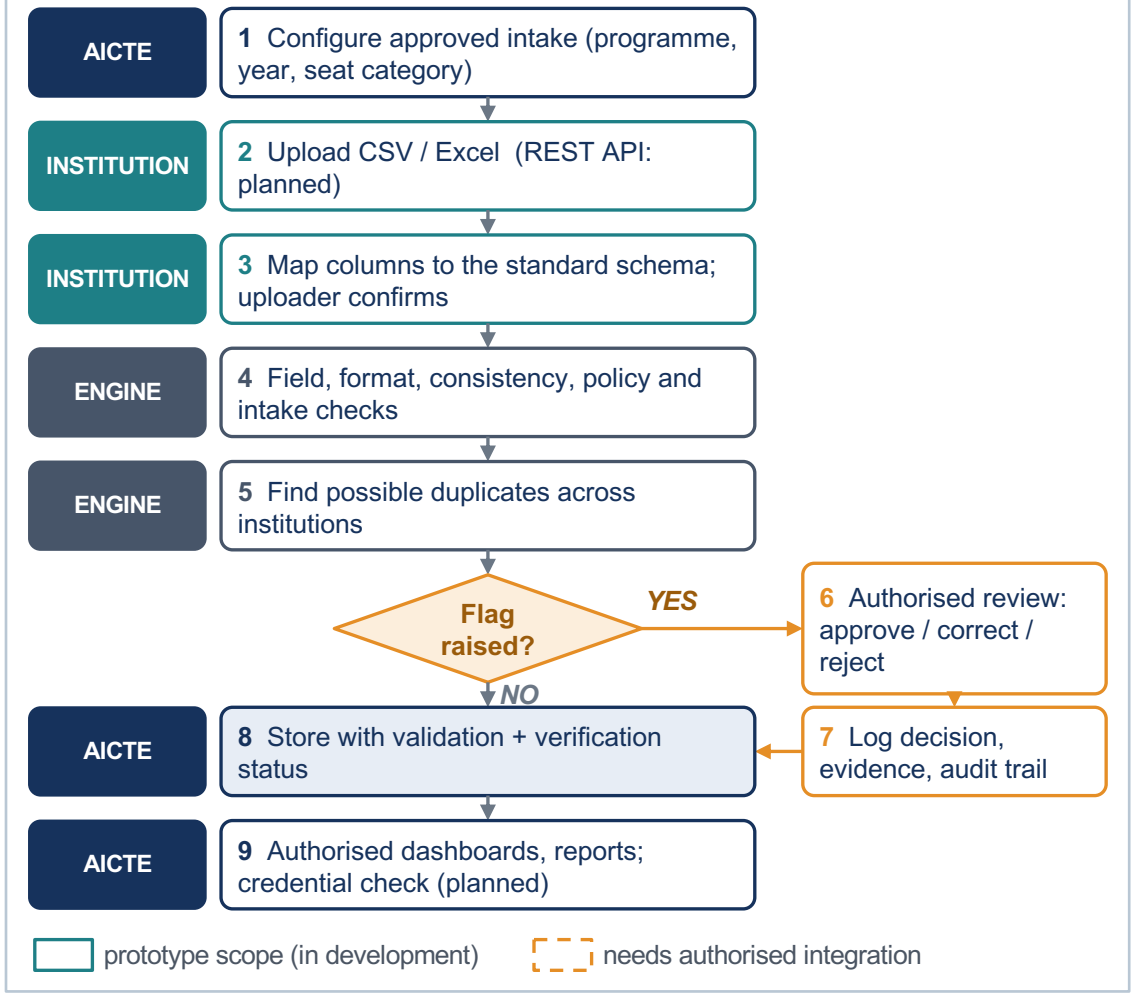
- **Explainable validation** — each flag names the field, rule and evidence
- **Institution-specific mapping** — a "PRN" column is mapped to the standard student_id field
- **Cross-institution duplicate candidates** — scored, reviewed, never auto-merged
- **Category-wise intake monitoring** — regular and supernumerary limits kept separate
- **Auditable review** — who decided what, when and why

EXAMPLE · INTAKE (illustrative)
B.Tech CSE, 2026-27
Approved regular intake: 120
Admitted on regular seats: 125
→ **Potential excess: 5 — sent for review**
Supernumerary seats (e.g. TFW, eligible NRI/OCI/FN) are checked against their own limits.

EXAMPLE · CONSISTENCY (illustrative)
Admitted 2026, 4-year programme
Expected graduation: 2030
Recorded graduation year: 2028
→ **Potential inconsistency — review**
Could be a data error, lateral entry or transfer; a reviewer decides.

END-TO-END WORKFLOW

FIGURE NO: 01



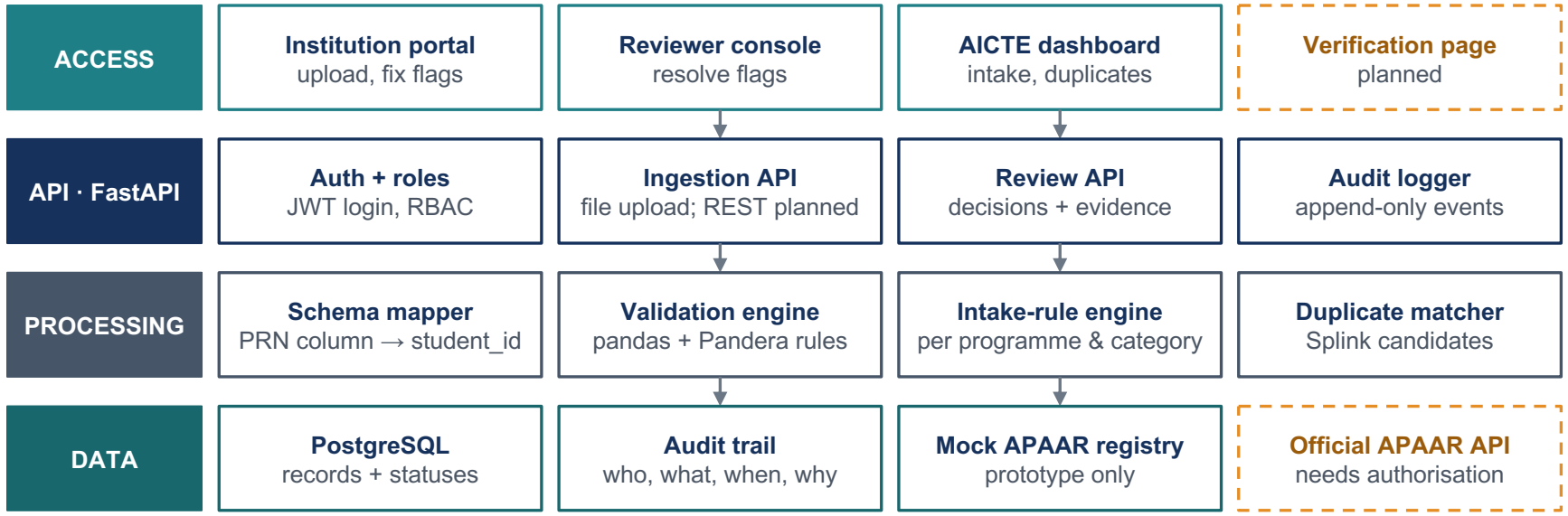
Team Name

TECHNICAL APPROACH

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SYSTEM ARCHITECTURE

FIGURE NO: 02



DUPLICATE-CANDIDATE MATCHING

1 Block — compare only records sharing DOB or a phonetic name key
2 Compare — name, DOB, gender, APAAR ID, institution
3 Score — weighted similarity
4 Route — score ≥ 0.75 → review queue (provisional)
5 Decide — reviewer marks same / different; never auto-merged

Thresholds will be calibrated on labelled test pairs. A high score means "check this pair", not proof of identity.

RECORD STATUS MODEL

Check	Possible status
Validation	passed · failed
Duplicate check	clear · potential duplicate
Review	not needed · pending · resolved
Official APAAR check	not checked · mock-checked · officially verified*

**only through an authorised integration. "Validation passed" is not official verification.*

TECH STACK

Frontend
React, TypeScript, Tailwind — portals and dashboards

Backend
FastAPI, Python — APIs and background jobs

Validation
pandas + Pandera — schema and rule checks

Matching
Splink on DuckDB — duplicate candidates

Database
PostgreSQL — records, statuses, audit trail

Security
JWT login, RBAC, audit log; Ed25519 signing (planned)

Deployment
Docker Compose — one-command setup

Team Name

FEASIBILITY AND VIABILITY

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FEASIBILITY

Technical

- ❖ Upload, mapping and validation work without any external integration
- ❖ Mature open-source libraries for rules and matching

Operational

- ❖ Institutions already keep admission lists in spreadsheets
- ❖ Needs trained reviewers at institution and AICTE level

Financial

- ❖ Open-source stack; a pilot can run on one server
- ❖ Hosting and support costs to be estimated with AICTE

RISKS AND HOW WE HANDLE THEM

Risk	Level	How we handle it
External APAAR / ABC dependency	High	API access, permissions, supported operations and credentials must be confirmed with the authorised service; the prototype uses a mock registry until then
Different column names and formats	High	Schema mapping confirmed by the uploader, plus a downloadable template
False duplicate matches (name variants)	Medium	Candidates only; a person decides; thresholds calibrated on labelled pairs
Wrong intake conclusions	Medium	Separate limits per seat category; excess shown as “potential” until reviewed
Personal data protection (DPDP Act, 2023)	High	Data minimisation, consent records, role-based access, masking, audit log; legal review before any real data
Reviewer workload at admission peaks	Medium	Review queue prioritised by confidence and impact

ROLLOUT PLAN

Phase 0 — Prototype
synthetic data, mock registry (current)

➡

Phase 1 — Pilot
volunteer institutions, subject to AICTE approval

➡

Phase 2 — Integration
official APAAR / ABC access, if authorised

➡

Phase 3 — Scale-up
wider rollout after pilot review

Team Name

IMPACT AND BENEFITS

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Impact on Each Stakeholder

Stakeholder	Impact	Benefit
AICTE	Student-level, APAAR-linked records per programme	Potential excess intake and duplicate candidates seen at submission time
Institutions	One structured upload with explained flags	Fewer clarification rounds with AICTE
Students	One consistent record linked to their APAAR ID	Fewer repeated document checks
Employers & Regulators	Check a credential's issuer signature and status (planned)	Faster than manual email or phone verification
Policy Makers	Programme- and category-level enrolment data	Evidence for seat planning and accreditation

How We Will Evaluate the Prototype

Rule checks recall on errors injected into synthetic batches	**Duplicate candidates** precision and recall on labelled pairs	**Review effort** flags per 1,000 records; time per flag	**Performance** processing time per 10,000 rows

No results are claimed yet; figures will be reported from the prototype.

Benefits

Social

- Fairer seat allocation: one student, one seat
- Consistent, portable academic record

Economic

- Less manual cross-checking
- Open-source stack, no licence fees

Environmental

- Fewer paper and couriered documents

Governance

- Audit trail for every change
- Consent-based data handling

Team Name

RESEARCH AND REFERENCES

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RESEARCH WE BUILT ON

[1] S. Schelter et al., "Automating Large-Scale Data Quality Verification," PVLDB 11(12), 2018. — *rule checks and metric history → our validation and drift checks*

https://www.vldb.org/pvldb/vol11/p1781-schelter.pdf

[2] P. Christen, Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection, Springer, 2012. — *blocking and field comparison → duplicate candidates*

https://link.springer.com/book/10.1007/978-3-642-31164-2

[3] dedupe library, API documentation. — *human-labelled pairs → review workflow and threshold calibration*

https://docs.dedupe.io/en/latest/API-documentation.html

[4] R. P. Wibowo et al., "Problem identification and intervention in the higher education data synchronization system in Indonesia," Procedia Computer Science 197, 2022. — *data problems in national institution reporting → problem framing*

https://doi.org/10.1016/j.procs.2021.12.165

[5] T. A. Kustitskaya et al., "Designing an Education Database in a Higher Education Institution for the Data-Driven Management of the Educational Process," Education Sciences 13(9), 947, 2023. — *student-record schema design → our standard schema*

https://doi.org/10.3390/educsci13090947

[6] J. A. Berrios Moya, J. Ayoade, M. A. Uddin, "A Zero-Knowledge Proof-Enabled Blockchain-Based Academic Record Verification System," Sensors 25(11), 3450, 2025. — *selective disclosure → future scope*

https://doi.org/10.3390/s25113450

DATASETS (SYNTHETIC ONLY)

student-performance (Hugging Face, CC0-1.0) — value ranges and missing-value tests

https://huggingface.co/datasets/neuralsorcerer/student-performance

UniversityIntake-MLDataset (GitHub, CC BY 4.0) — applicant-stage field tests

https://github.com/drsanjayagal/UniversityIntake-MLDataset

Neither dataset contains APAAR IDs; test IDs are generated. No real student data is used.

STANDARDS REFERENCED

- **W3C Verifiable Credentials Data Model 2.0** (W3C Recommendation, May 2025) — planned credential format
- **RFC 8032 (EdDSA / Ed25519)** — planned record signing
- **Digital Personal Data Protection Act, 2023** — privacy design input
- **AICTE Approval Process Handbook** — intake rules (current edition to confirm)